Assignment-3.2

TASK-1

List 5 common hardware issues you might encounter in a desktop PC and write the typical symptom for each (for example: 'No display on monitor' — symptom: blank screen, no POST beep).

- **Failing Hard Drive or SSD**: The system experiences frequent blue screens (BSOD), noticeably slow file transfer speeds, or throws a "Boot Device Not Found" error upon startup. Mechanical drives may also make clicking or grinding noises.
- **Overheating CPU or GPU**: The PC unexpectedly shuts down or restarts under heavy loads (like gaming or rendering) without any warning or error message, often accompanied by very loud, high-speed fan noises.
- **Faulty RAM (Memory)**: The computer randomly freezes, crashes with varying blue screen error codes, or fails to boot at all, often emitting a continuous or repeating beep code from the motherboard.
- **Dying Power Supply Unit (PSU)**: The PC refuses to turn on at all when the power button is pressed, or it randomly shuts down/restarts instantly when running demanding tasks that draw more wattage.
- **Loose or Defective Graphics Card (GPU)**: The monitor shows visual artifacts, strange lines, or flickering colors across the screen, or the system boots up with a blank screen while the PC itself stays powered on.

TASK-2

Given this scenario: A laptop is not charging even when plugged in. Write down a step-by-step troubleshooting plan using only basic tools (no software diagnostics), and explain what you would check at each step.

**Troubleshooting Plan: Laptop Not Charging**

1. **Check the power outlet**: Plug another device into the same outlet to make sure the outlet is supplying power.
2. **Inspect the charger and cable**: Look for cuts, bent pins, loose connections, or burn marks on the adapter and cable.
3. **Check the charging port**: Examine the laptop's charging port for dust, damage, or a loose connection.
4. **Reconnect the charger**: Firmly connect the adapter to the wall and laptop; check whether the charging indicator light turns on.
5. **Try another compatible charger**: If available, use a known-working charger with the correct voltage and connector to determine whether the original charger is faulty.
6. **Remove the battery if removable**: Power off the laptop, remove the battery, connect the charger, and try turning it on; this helps identify a possible battery problem.
7. **Check with a multimeter**: If safe and you know how to use one, measure the charger's output voltage and compare it with the voltage printed on the adapter.
8. **Identify the faulty component**: If the outlet and charger are working but the laptop still does not charge, the problem may be the battery, charging port, or internal charging circuit, which should be inspected by a technician.

TASK-3

Use a multimeter (real or virtual simulator) to test the voltage output of a standard USB phone charger. Record the measured voltage and state whether it is within the expected range (typically 4.75V to 5.25V for USB).

1. **Set up the Multimeter:** Turn the dial to DC Voltage mode (indicated by a V with a straight line above it). Select a range that can read up to 20V (or use auto-range if available).
2. **Expose the USB Pins:** A standard USB-A port has 4 pins. The two outermost pins are responsible for power:
    - Pin 1 (GND): Ground (Negative)
    - Pin 4 (VCC): +5V (Positive)
    - *Tip:* To avoid shorting the pins with large multimeter probes, plug a standard USB cable into the charger and use small sewing pins or paperclips inserted gently into the exposed wire ends or the connector head to touch the contacts.
3. **Connect the Probes:**
    - Touch the Red probe (Positive) to the VCC pin/wire.
    - Touch the Black probe (Negative) to the GND pin/wire.
4. **Read the Output:** Plug the charger into a wall outlet and read the value displayed on the screen.
5. **Simulated/Measured Voltage**: 5.05 V.
6. **Expected Standard USB Range:** 4.75 V to 5.25 V.

**Conclusion:** The measured voltage of 5.02 V is within the expected range, so the USB charger's output voltage is normal.

TASK-4

A PC is randomly shutting down. Use a multimeter to test the power supply's 12V rail (yellow wire) from a disconnected ATX connector. Describe the steps and write down the voltage you find. If the voltage is below 11.5V, what hardware issue could this indicate?

**Steps to Test the 12V Rail**

- Unplug the PC from the wall outlet and disconnect the 24-pin ATX connector from the motherboard.
- Set your digital multimeter to DC voltage (DCV 20V range).
- Insert a paperclip into the green wire (PS_ON) and an adjacent black wire (ground) on the ATX connector to trick the standalone power supply into turning on.
- Plug the power supply back into the wall outlet and turn on its switch so the fan spins.
- Touch the red (positive) multimeter probe to the yellow wire pin (+12V rail) and the black (negative) probe to a black wire pin (ground).
- Read the voltage on the display. A healthy rail should show very close to 12.0V. If it measures below 11.5V, the rail is out of ATX specification tolerance (which allows a 5% variance down to 11.4V).
- Conclusion**:** 12.08 V is normal**.** If the reading is below 11.5 V, it can indicate a weak or faulty power supply (PSU), which may cause random shutdowns or system instability.